Computer Architecture (PCC-CS402)

Course Details

Credit Points: 3

Pre-Requisites

1. Basic Structure of Computers, Functional units, software, performance issues software, machine instructions

2. RAM, ROM, Memory management

Syllabus Outline

Unit 1

- **Introduction**: Review of basic computer architecture (Revisited), Quantitative techniques in computer design, measuring and reporting performance.
- **Pipelining**: Basic concepts, instruction and arithmetic pipeline, data hazards, control hazards and structural hazards, techniques for handling hazards. Exception handling. Pipeline optimization techniques; Compiler techniques for improving performance.

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Unit 2

Hierarchical memory technology

- Inclusion, Coherence and locality properties; Cache memory organizations, Techniques for reducing cache misses; Virtual memory organization, mapping and management techniques, memory replacement policies.

Unit 3

Instruction-level parallelism

- Basic concepts, techniques for increasing ILP, superscalar, super-pipelined and VLIW processor architectures. Array and vector processors.

Unit 4

- Multiprocessor architecture: taxonomy of parallel architectures
- Centralized shared- memory architecture: synchronization, memory consistency, interconnection networks. Distributed shared-memory architecture. Cluster computers.
- Non von Neumann architectures: data flow computers, reduction computer architectures, systolic architectures.

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Computer Architecture Lab (PCC-CS492)

Credit Points: 2

Laboratory Experiments

1. HDL introduction.
2. Basic digital logic base programming with HDL
3. 8-bit Addition, Multiplication, Division
4. 8-bit Register design
5. Memory unit design and perform memory operations.
6. 8-bit simple ALU design
7. 8-bit simple CPU design
8. Interfacing of CPU and Memory.